

SIH 2026 Presentation Kit: Air Power

Oct 9, 2026 · @Blah Blah

Every number below comes from our own runs on the 100 unseen NASA FD001 test engines. Quote these exact figures and nothing else.

Key numbers (verified from our runs)

Measure	Value	Note
Data	NASA C-MAPSS FD001: 100 training engines (20,631 rows), 100 test engines	Public and simulated, not real Air Force data
Model	Random Forest, 200 trees, 14 sensors, 5-flight smoothing, answer capped at 125 flights	Trained only on the 100 training engines
Average error (MAE)	13.1 flights (vs capped truth), 14.2 flights (vs raw truth)	Lazy "always the average" guess: 34.8 flights (vs capped truth)
RMSE	18.5 flights (vs capped truth), 19.5 flights (vs raw truth)	Lazy guess: 41.9 flights (vs capped truth)
Fleet status	62 Green, 16 Yellow, 22 Red	Ready tomorrow (Green + Yellow): 78 of 100
Red alerts that caught truly at-risk engines	21 of 25 engines within 30 flights of failure; 4 missed; 1 false alarm	With the 10-flight safety margin. Without it: 17 of 25 caught
Dangerous misses	17 of 100 engines the model guessed 20+ flights too optimistic	Reason for the safety margin
Sensor drift (average of the biggest drift per aircraft)	Green about 2, Yellow about 6, Red about 10 noise units	Backs the colours with real sensor behaviour

Measure	Value	Note
Backtest, 7 days: surprise failures	16 if we do nothing, 10 with our plan	Forecast (using our estimates): 18 vs 14
What-if: 3 extra spares per base	Surprise failures fall from 16 to 5 in the backtest	Check live with the slider before presenting

What is simulated by us (say this out loud)

- Bases (random assignment, fixed seed 42), spare stock (3, 1 and 0 modules at Base A, B, C), resupply time (2, 4 and 6 days) and free hangar slots.
- Assumptions: 3 flights per aircraft per day, 2 hangar days per maintenance, 3 extra days lost for a surprise failure, priority score = 70% risk + 30% spares shortage, 10-flight safety margin.
- The 10-flight margin was chosen after looking at test results. Say so if asked.

6-slide PPT content

Our pitch in one line: most teams predict which engine will fail, we also tell the commander what to fix first, given spare parts and hangar slots. Use a light background and the dashboard's colours (blue, green, amber, red) so the slides match the demo. Text in [brackets] is for you to fill in.

Slide 1: Title

- **Air Power: Predictive Maintenance & Fleet Availability**
- Which aircraft to fix first, why, and how many will be ready this week
- Smart India Hackathon 2026 | PS 26249 | Ministry of Defence (Defence Services Staff College) | Software | Transportation & Logistics
- [Team name] | [Member names] | Sapthagiri College of Engineering
- Code: github.com/jeevikagowda7/sih26-airpower-predictive-maintenance
- Visual: screenshot of the dashboard's top five cards.

Slide 2: Idea

- **Problem:** aircraft availability is low because maintenance is mostly reactive, and health data, technical records, spares and maintenance slots sit in separate places.

- **Our idea:** one dashboard that joins engine health, spares and hangar slots, and answers "what should we fix first, and when?"
- **What makes it different:**
 1. Maintenance priority score: failure risk plus spares shortage, with a booked slot for each aircraft.
 2. "Why is it red?": the sensors that drifted most from the aircraft's own early life.
 3. 7-day availability forecast with what-if sliders (spares delayed, slots lost, extra spares).
- Visual: a simple left-to-right flow: Sensor data → AI estimate of flights left → Green/Yellow/Red → Priority score (with spares and slots) → 7-day forecast.
- Say: "Predicting a failure is not enough. A commander needs to know which aircraft to fix first when the spare part and the hangar are limited."

Slide 3: Technical approach

- **Data:** NASA C-MAPSS FD001 (public, simulated turbofan data): 100 engines run to failure for training, 100 unseen engines for testing. Spares, bases and slots are simulated by us and labelled as such.
- **Model:** Random Forest estimates Remaining Useful Life (flights left) from 14 useful sensors, smoothed over 5 flights.
- **Status:** Red at 30 flights or fewer, Yellow at 31 to 60, Green above 60, after a 10-flight safety margin.
- **Explanation:** each sensor's drift from that aircraft's own first 20 flights, in units of normal noise.
- **Planner:** gives spares and hangar slots to the riskiest aircraft first, then simulates the next 7 days.
- **Stack:** Python, pandas, scikit-learn, Streamlit, Plotly, GitHub.
- **Measured results (100 unseen engines):** average error 14.2 flights (a lazy guess is off by about 35); 21 of 25 truly at-risk engines flagged Red, 1 false alarm.
- Visual: pipeline diagram plus a small results box.

Slide 4: Feasibility and viability

- **Feasible today:** a working prototype exists, runs on a normal laptop, uses only open-source tools, and anyone can reproduce it from our GitHub repository.

- **Path to real data:** the dashboard reads plain tables (engine health, spares stock, slot availability). Real health-monitoring, technical-record and spares data can replace our simulated inputs. We have not tested real Air Force data.
- **Viable to run:** no licence cost; can run on-premises, which suits restricted networks.
- **Risks and how we handle them:**

Risk	Mitigation
Real data is messier than NASA's	Pilot on one fleet's historical data first; retrain on it
Model too optimistic (17 of 100 test engines)	Safety margin; always show the uncertainty to the engineer
False alarms or missed faults	Report both rates openly; engineers make the final call
Data security	On-premises deployment; access control

- **Proposed roadmap:** Phase 1 pilot on historical data of one fleet; Phase 2 connect to maintenance and spares systems; Phase 3 extend to more fleets and fault types.

Slide 5: Impact and benefits

- **Fewer surprise failures:** in our simulated 7-day backtest, 16 surprise failures with no planning fall to 10 with our plan, and to 5 when the what-if adds 3 spares per base.
- **Better decisions:** shows spares shortages and missing slots before an aircraft is grounded, and flags aircraft whose booked slot comes after their likely failure.
- **Higher readiness over time:** fewer emergency repairs means more planned, shorter downtime. Within the first week, planned maintenance takes aircraft out early, so the real gain is fewer breakdowns.
- **Safer operations:** earlier warning for engines drifting towards failure.
- **Reusable:** the same approach applies to other fleets such as helicopters, vehicles and civil aircraft.
- Footnote on the slide: "Figures come from simulated data and stated assumptions. Real impact must be measured in a pilot."
- Visual: screenshot of the 7-day forecast and the surprise-failures cards.

Slide 6: Research and references

- A. Saxena, K. Goebel, D. Simon, N. Eklund, "Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation", International Conference on Prognostics and Health Management (PHM08), Denver, 2008.
- NASA Prognostics Center of Excellence Data Set Repository: C-MAPSS turbofan engine degradation simulation data.
- L. Breiman, "Random Forests", Machine Learning 45(1), 5 to 32, 2001.
- F. Pedregosa et al., "Scikit-learn: Machine Learning in Python", Journal of Machine Learning Research 12, 2825 to 2830, 2011.
- Tools: Streamlit, Plotly, pandas.
- Future work: harder NASA sets (FD002 to FD004), deep-learning models, SHAP-style explanations, real maintenance records.
- Code and live prototype: github.com/jeevikagowda7/sih26-airpower-predictive-maintenance

2-minute demo script

The script is about 260 spoken words, which fits two minutes at a calm pace. Practise it three times with a timer, and record a backup screen video in case the laptop misbehaves on the day. Before you start: open the dashboard at 100% browser zoom, sliders at their defaults, and the Fleet table tab showing.

Time	What to show	What to say
0:00 to 0:15	Top of the page: banner and the five cards	"This is our fleet of 100 aircraft. 62 are healthy, 16 need watching and 22 are red. 78 are ready tomorrow. The engine data is NASA's public simulated dataset, and our spares and hangar slots are simulated by us. The page says so."

Time	What to show	What to say
0:15 to 0:40	Fleet table, scroll the first rows (AC-035)	"For each aircraft we show the AI's estimate of flights left, the main reason, the spares position, and what to do. Look at AC-035: no spare at its base until day 6, and its booked slot may come after it fails. A plain colour chart would never show that."
0:40 to 1:00	Alerts and priorities tab	"Among the red aircraft, 18 are short of spares, 11 have no hangar slot in the next 7 days, and 8 have a slot after their likely failure. The priority score combines failure risk with this supply problem, so the commander sees what to fix first."
1:00 to 1:25	Aircraft detail tab, choose AC-035, show the drift bars	"Why is it red? Core speed rose 7 times its normal noise, corrected core speed 6.8 times, and compressor pressure 5 times. These are compressor sensors, which matches the compressor fault in this dataset. It is a pointer for engineers, not proof."
1:25 to 1:50	7-day forecast tab; move 'Extra spare modules per base' to 3, then back to 0	"Now the what-if. Replaying our plan against NASA's true answers, surprise failures fall from 16 to 10. If we add three spares per base, they fall to 5. So the tool shows where one more spare part helps the most."
1:50 to 2:00	Simulated data and accuracy tab	"Our average error is 14 flights, against 35 for a lazy guess. We flagged 21 of 25 at-risk engines with 1 false alarm, and we missed 4. We show that openly. The code is on GitHub. Thank you."

If something goes wrong: say "let me show the recording" and play the backup video. Do not apologise for long.

Do not say: "real Air Force data", "100% accurate", "fully explainable AI" or "digital twin". Say "digital-twin-style what-if simulator" if you need the term.

The 10 hardest judge questions

Every answer below is honest and uses only numbers we measured. Say them in your own words, and keep each to about 20 seconds.

1. **Is this real Air Force data?** No. The engine data is NASA's public C-MAPSS simulation, and the bases, spares and hangar slots are simulated by us. All of it is labelled on the dashboard. The dashboard reads plain tables, so real data can replace ours, but we have not tested that.
2. **How accurate is your model?** On 100 engines it had never seen, the average error is 14.2 flights, against about 35 for a lazy "always the average" guess. It flagged 21 of 25 truly at-risk engines as Red, with 1 false alarm and 4 missed. It was 20 or more flights too optimistic on 17 of 100 engines, and we show that openly.
3. **Why a Random Forest and not deep learning?** It trains in seconds on small data, needs no special hardware, and is easy to explain to engineers. We tested only this model against a lazy baseline, so we do not claim it is the best. Deep-learning models are on our roadmap.
4. **What happens when the model is wrong, and how did you choose your thresholds?** Red at 30 flights and Yellow at 60 are our policy settings, and engineers make the final call. We added a 10-flight safety margin because the model was too optimistic. We chose that margin after looking at test results, so in real use it must be set on separate validation data.
5. **What is new compared with existing predictive-maintenance tools?** The individual parts are standard. What we add is joining engine health with spares and hangar slots into one priority list with booked slots, a warning when a slot comes after the likely failure, and a what-if simulator. It is decision support, not just a failure alarm.
6. **Is your "why is it red" real explainable AI?** It is a simple sensor-drift explanation, not SHAP. It shows which sensors moved most from that aircraft's own early life. Across the fleet the biggest drift averages about 2 noise units for Green, 6 for Yellow and 10 for Red. It is a pointer for engineers, not proof of cause.
7. **Real fleets have many aircraft types, conditions and faults. Does this work for them?** Our data has one flying condition and one fault type, compressor degradation. NASA's harder sets have six conditions and two faults, and we have not trained on them. A real deployment would retrain on each fleet's own data.

8. **How would it connect to real maintenance and health-monitoring systems?** Today it reads three tables: engine health, spares stock and slot availability. A real system needs data feeds from the actual systems, with agreed formats and approvals. We have not built that, and it is Phase 2 of our roadmap.
9. **What about security and deployment on defence networks?** The prototype runs on a single computer with no cloud service, so it can run on-premises and offline. We have not built user login, audit logs or access control yet. Those would be needed before any real use.
10. **How do you prove it improves aircraft availability?** In simulation only. Replaying our plan against NASA's true answers, surprise failures in 7 days fall from 16 to 10. Availability during that first week is not higher, because planned maintenance takes aircraft out early. A pilot on historical records should measure unplanned downtime, spares stock-outs and availability percentage.

Idea description for the SIH portal

Paste the full version first. If the portal has a character limit, use the short version. Check the portal's field names and limits yourself before pasting, because they can differ.

Title: Air Power: Predictive Maintenance & Fleet Availability Platform

Full version (265 words, about 1,700 characters)

Low aircraft availability is driven by reactive maintenance and by data that sits in separate places: health monitoring, technical records, spares and maintenance slots. Air Power is an AI-based decision-support platform that joins these into one view and tells the commander which aircraft to fix first, why, and how many will be ready this week.

A machine-learning model estimates each aircraft's Remaining Useful Life (flights left) from engine sensor data and turns it into a Green, Yellow or Red status. A scheduler then combines failure risk with spare-part stock and free hangar slots to produce a maintenance priority score and a booked slot for each aircraft, and warns when a slot comes after the likely failure. For every Red aircraft, the system shows which sensors drifted most from that aircraft's own early-life behaviour, as a pointer for engineers. A 7-day availability forecast with what-if sliders (spares delayed, extra spares, hangar slots lost) works as a simple digital-twin-style simulator.

We built a working prototype on NASA's public, simulated C-MAPSS turbofan data (FD001). On 100 unseen engines the average error is 14.2 flights, against about 35 for a naive guess, and 21 of 25 truly at-risk engines were flagged Red with 1 false alarm. Spares,

bases and hangar slots in the prototype are simulated by us and clearly labelled. In a simulated 7-day backtest, surprise failures fell from 16 to 10.

The system uses open-source tools, runs on a single computer, and can run on-premises. It reads plain data tables, so real health-monitoring, technical-record and spares data can replace our simulated inputs in a pilot. Code: github.com/jeevikagowda7/sih26-airpower-predictive-maintenance

Short version (97 words, about 600 characters)

Air Power is an AI decision-support platform that joins aircraft health data, spares stock and hangar slots to tell commanders which aircraft to maintain first and how many will be ready this week. A Random Forest model estimates flights left for each aircraft and sets Green, Yellow or Red status. A scheduler adds a priority score, booked slots, a "why is it red" sensor explanation and a 7-day what-if forecast. Prototype on NASA's public simulated data: average error 14.2 flights, 21 of 25 at-risk engines flagged. Spares and slots are simulated and labelled. Open source, runs on-premises.

Technologies: Python, pandas, scikit-learn (Random Forest), Streamlit, Plotly, Git and GitHub.

Dataset used: NASA C-MAPSS turbofan engine degradation simulation data (FD001), public and simulated, plus simulated spares and maintenance-slot tables created by the team.